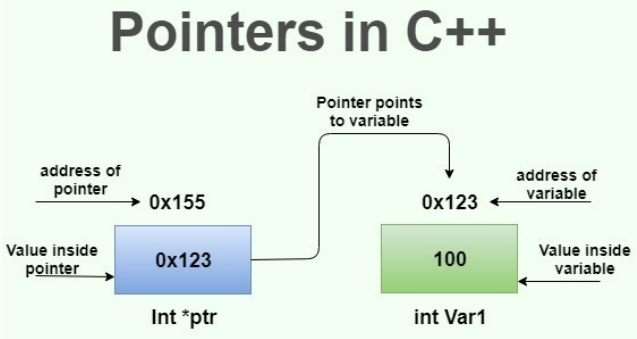
C++ Fundamentals - Part 4

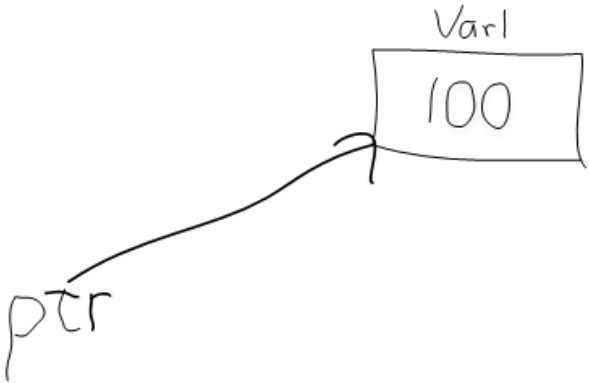
1. Drawing Pointers
2. Pointers Versus References

Drawing Pointers



Source: https://simplesnippets.tech/cpp-pointers-concept-with-example/

Simplified Conceptual Drawing



Pointers and References

Pointers hold the address of an object.
References provide an *alias* for an object.
Both are forms of indirection.

```
// construct a string object
string name("Bob");

// pointer to the string object
string* string_ptr = &name;
cout << *string_ptr << endl;

// a reference, or alias, for the string object
string& string_ref = name;
cout << string_ref << endl;
```

Pointers and References

Sketch a conceptual drawing of the following:

```
string name("Bob");

string* string_ptr = &name;

string& string_ref = name;

string** string_ptr_ptr = &string_ptr;
```

Is a Pointer Just a Reference in Disguise?

Is a Pointer Just a Reference in Disguise?

A pointer is a separate object (from the pointee) with its own distinct set of operations, lifetime, size, and memory location

```
// construct a string object
string name("Bob");
string* string_ptr = &name;

// dereferencing, incrementing
// only valid for ptrs - not string
*string_ptr;  // *name;  <- error
string_ptr++; // name++; <- error
```

Operations for a reference are defined by what it is referring to:

```
string& string_ref = name;
// clear() is a member function of string
string_ref.clear();
```

A pointer can point to null but a reference has to alias an *existing* object.